Math 110 — Week 3

Linear maps, null space and range, matrices (Notes Ch. 5–6, LADR 3A–3C, HW3)

§1 Facts used from Weeks 1–2

(W1) Subspace test: $0 \in U$ and $\lambda v + w \in U$. Direct sum: every vector of $U_1 \oplus \cdots \oplus U_m$ is $u_1 + \cdots + u_m$ in exactly one way (Week 1, Prop. 7).

(W2) $v_1, \dots, v_n$ is a basis of V exactly when every $v \in V$ is $\lambda_1 v_1 + \cdots + \lambda_n v_n$ in exactly one way (Week 2).

(W3) An independent list extends to a basis; a spanning list contains a basis (Week 2, Props. 9, 11).

(W4) In a space of dimension d: d independent vectors form a basis; a subspace of dimension d is the whole space (Week 2, Props. 13, 14).

(W5) Pigeonhole: a map between finite sets of equal size is injective exactly when it is surjective (Week 1).

§2 This week

§2.1 Linear maps (Ch. 5.1)

The question. Which functions between vector spaces respect the two operations? For such a function, how much information is needed to know it completely?

- Differentiation: $(\lambda p + q)' = \lambda p' + q'$. Knowing the derivatives of $1, x, x^2, x^3$ gives the derivative of every cubic.

Definition (Def. 11, 12 / LADR 3.1, 5.1)**.** $T\colon V \to W$ is **linear** if $T(\lambda v + w) = \lambda T(v) + T(w)$ for all $v, w \in V$, $\lambda \in \mathbb{F}$. Equivalently $T(v + w) = Tv + Tw$ and $T(\lambda v) = \lambda Tv$; equivalently $T(\sum \lambda_k v_k) = \sum \lambda_k T(v_k)$. The set of linear maps $V \to W$ is $\mathcal{L}(V, W)$. A linear map $V \to V$ is an **operator**; $\mathcal{L}(V) = \mathcal{L}(V, V)$.

- Always $T(0) = 0$: $T(0) = T(0 \cdot 0) = 0 \cdot T(0) = 0$. So a map with $T(0) \neq 0$ is not linear.
- Ex. 21: matrix maps, differentiation, the map from a periodic function to its Fourier coefficients.
- Ex. 22: the **shift** $S(a_0, a_1, a_2, \dots) = (a_1, a_2, a_3, \dots)$ on $\mathbb{R}^{\mathbb{N}_0}$. It maps the recurrence subspace $R = \{a : a_{n+2} = a_{n+1} + a_n\}$ (Week 1) into itself, and on R it satisfies $S^2 - S - I = 0$, because $(S^2 a)_n = a_{n+2} = a_{n+1} + a_n = (Sa + a)_n$.
- Ex. 23: three maps on $\mathbb{R}^2$ used all semester. Projection $P(x, y) = (0, y)$. Rotation $R_\theta(x, y) = (x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta)$. And $N(x, y) = (y, 0)$, for which $Ne_2 = e_1$, $Ne_1 = 0$, so $N^2 = 0$. Also $N = R_{-\pi/2} P$.

Proposition (Prop. 18, 19 / LADR 3.6–3.8)

$\mathcal{L}(V,W)$ is a vector space with $(\mu S+T)(v)=\mu S(v)+T(v)$. If $T\in\mathcal{L}(U,V)$ and $S\in\mathcal{L}(V,W)$, the composition ST, $(ST)(u)=S(T(u))$, is linear.

Proof. 1. For S,T linear and scalars λ,μ, linearity of S and of T gives

$$(\mu S+T)(\lambda v+w)=\mu(\lambda Sv+Sw)+\lambda Tv+Tw=\lambda(\mu S+T)(v)+(\mu S+T)(w),$$

so $\mu S+T$ is linear. The zero map is linear. So $\mathcal{L}(V,W)$ is a subspace of the space of all functions $V\to W$.

2. Apply the linearity of T, then of S:

$$ST(\lambda u+v)=S(\lambda Tu+Tv)=\lambda STu+STv.$$

□

- Composition is associative, $TI=T=IT$, and distributive: $(S_1+S_2)T=S_1T+S_2T$, $S(T_1+T_2)=ST_1+ST_2$.
- It is not commutative. Ex. 24: on smooth functions let $(\hat{X}f)(x)=xf(x)$ and $\hat{P}f=f'$. Then $(\hat{P}\hat{X}-\hat{X}\hat{P})f=(xf)'-xf'=f$, so $\hat{P}\hat{X}-\hat{X}\hat{P}=I$.
- Def. 13: the **restriction** of T to a subspace U is $T|_U\colon U\to W$, $T|_U(u)=T(u)$.

Theorem (Prop. 21 / LADR 3.4: a linear map is determined by its values on a basis)

Let $v_1,\dots,v_n$ be a basis of V and $w_1,\dots,w_n$ any vectors in W. There is exactly one linear map $T\colon V\to W$ with $T(v_j)=w_j$ for all j.

The idea

Uniqueness comes first and tells us what the map must be. Every v is $\lambda_1v_1+\cdots+\lambda_nv_n$, so linearity forces $T(v)=\lambda_1w_1+\cdots+\lambda_nw_n$. There is no choice.

For existence, take that formula as the definition. Is it allowed? The danger with "define $T(v)$ using a way of writing v" is that v might be written in two ways that give different answers. Here it cannot: the coefficients λ_k are unique because $v_1,\dots,v_n$ is a basis (W2). That is the only place the basis property is used. If the list were dependent the definition could contradict itself; if it did not span, T would not be defined everywhere.

Proof. 1. *Uniqueness.* If T is linear with $T(v_j)=w_j$ and $v=\sum\lambda_jv_j$, then $T(v)=\sum\lambda_jw_j$.

2. *Existence.* For $v=\sum\lambda_jv_j$ (unique coefficients, W2) define $T(v)=\sum\lambda_jw_j$. Then $T(v_j)=w_j$.

3. *Linear.* If $v=\sum\lambda_jv_j$ and $v'=\sum\lambda'_jv_j$, then $\mu v+v'=\sum(\mu\lambda_j+\lambda'_j)v_j$, so

$$T(\mu v+v')=\sum(\mu\lambda_j+\lambda'_j)w_j=\mu T(v)+T(v').$$

□

Proposition (Prop. 20: gluing maps on a direct sum)

If $U_1\oplus\cdots\oplus U_m$ is a direct sum and $T_i\colon U_i\to W$ are linear, there is exactly one linear $T\colon U_1\oplus\cdots\oplus U_m\to W$ with $T|_{U_i}=T_i$ for all i: $T(u_1+\cdots+u_m)=T_1(u_1)+\cdots+T_m(u_m)$.

Proof. 1. *Uniqueness.* If T is linear and equals T_i on each U_i, then

$$T(u_1+\cdots+u_m)=T(u_1)+\cdots+T(u_m)=T_1(u_1)+\cdots+T_m(u_m).$$

So the formula is the only possibility.

2. *The formula is a valid definition.* Every vector of the direct sum is $u_1+\cdots+u_m$ with $u_i\in U_i$ in exactly one way (W1), so the formula assigns it exactly one value.

3. *Linear.* If $v = \sum u_i$ and $v' = \sum u_i'$, then $\lambda v + v' = \sum(\lambda u_i + u_i')$ with $\lambda u_i + u_i' \in U_i$, so

$$T(\lambda v + v') = \sum T_i(\lambda u_i + u_i') = \lambda \sum T_i(u_i) + \sum T_i(u_i') = \lambda T(v) + T(v').$$

4. *Restrictions.* A vector $u \in U_i$ is the sum with u in position i and 0 elsewhere, so $T(u) = T_i(u)$.

Prop. 21 is the special case where U_j is the line $\mathbb{F}v_j$. □

§2.2 Null space and range (Ch. 5.2)

The question. A linear map can lose information (send different vectors to the same place) and can miss part of the target. How are the two measured, and how are they related? For $P(x, y) = (0, y)$: everything on the x-axis is sent to 0, and only the y-axis is reached.

Definition (Def. 14, 15 / LADR 3.11, 3.16)**.** For $T \in \mathcal{L}(V, W)$: the **null space** (kernel) is $\operatorname{null} T = \{v \in V : Tv = 0\}$; the **range** (image) is $\operatorname{range} T = \{Tv : v \in V\}$. $\dim \operatorname{null} T$ is the **nullity**, $\dim \operatorname{range} T$ the **rank**.

Proposition (Prop. 22, 23 / LADR 3.13, 3.15, 3.18, 3.19)

$\operatorname{null} T$ is a subspace of V and $\operatorname{range} T$ is a subspace of W. T is injective $\iff$ $\operatorname{null} T = \{0\}$. T is surjective $\iff$ $\operatorname{range} T = W$.

The idea

For a general function, injectivity has to be checked for every pair of inputs. For a linear map, $Tv = Tw$ is the same as $T(v - w) = 0$, so two inputs collide exactly when their difference is in the null space. Checking the single equation $Tv = 0$ is enough.

Proof. 1. $T0 = 0$, so $0 \in \operatorname{null} T$. If $Tv_1 = Tv_2 = 0$ then $T(\lambda v_1 + v_2) = 0$.

2. $0 = T0 \in \operatorname{range} T$. If $w_1 = Tv_1$, $w_2 = Tv_2$ then $\lambda w_1 + w_2 = T(\lambda v_1 + v_2)$.

3. If T is injective and $Tv = 0 = T0$, then $v = 0$. Conversely, if $\operatorname{null} T = \{0\}$ and $Tv = Tw$, then $T(v - w) = 0$, so $v - w = 0$.

4. Surjective means every w is some Tv, which is $\operatorname{range} T = W$.

□

Ex. 25: D on $\mathcal{P}_4(\mathbb{R})$. $\operatorname{null} D = \operatorname{span}(1)$. $\operatorname{range} D$ is spanned by $D1, Dx, \ldots, Dx^4 = 0, 1, 2x, 3x^2, 4x^3$, so $\operatorname{range} D = \mathcal{P}_3(\mathbb{R})$.

Theorem (Prop. 24 / LADR 3.21: fundamental theorem of linear maps (rank–nullity))

Let V be finite-dimensional and $T \in \mathcal{L}(V, W)$. There is a basis $u_1, \ldots, u_m, v_1, \ldots, v_n$ of V such that $u_1, \ldots, u_m$ is a basis of $\operatorname{null} T$ and $Tv_1, \ldots, Tv_n$ is a basis of $\operatorname{range} T$. In particular

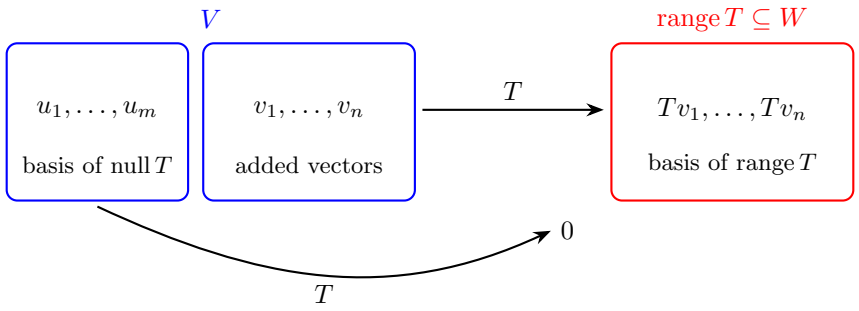
$$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T.$$

The idea

The range is spanned by the images of any basis of V. Some of those images may be 0 or redundant. To get an honest basis of the range, choose the basis of V so that the wasted vectors are separated out: start with a basis $u_1, \ldots, u_m$ of $\operatorname{null} T$ (their images are all 0) and extend it by $v_1, \ldots, v_n$.

Then $\operatorname{range} T$ is spanned by $Tv_1, \ldots, Tv_n$ alone. Are these independent? A relation $\sum \lambda_j T v_j = 0$ says $T(\sum \lambda_j v_j) = 0$, i.e. $\sum \lambda_j v_j$ is in the null space, i.e. it is a combination of the u's. But the u's and v's together are independent, so that is only possible with all $\lambda_j = 0$. In words: the added vectors v_j point in directions that T does not send to 0, and no combination of them is sent to 0 either.

For D on $\mathcal{P}_4(\mathbb{R})$: null space basis 1; extend by x, x^2, x^3, x^4; their images $1, 2x, 3x^2, 4x^3$ are a basis of the range. $5 = 1 + 4$.



A basis of V adapted to T. The first group is sent to 0. The images of the second group are a basis of the range. Counting: $\dim V = m + n$.

Proof. 1. $\operatorname{null} T$ is finite-dimensional (W4). Choose a basis $u_1, \ldots, u_m$ of it and extend to a basis $u_1, \ldots, u_m, v_1, \ldots, v_n$ of V (W3).

2. $Tv_1, \ldots, Tv_n$ *spans* $\operatorname{range} T$. Write any $v \in V$ as $v = \sum \mu_i u_i + \sum \lambda_j v_j$. Since every $Tu_i = 0$,
$$Tv = \sum \mu_i T u_i + \sum \lambda_j T v_j = \sum \lambda_j T v_j.$$

3. *Independent.* Suppose $\sum \lambda_j T v_j = 0$. Then $T(\sum \lambda_j v_j) = 0$, so $\sum \lambda_j v_j \in \operatorname{null} T$, and $\sum \lambda_j v_j = \sum \mu_i u_i$ for some μ_i.

4. Then $\sum \mu_i u_i - \sum \lambda_j v_j = 0$ is a relation in a basis of V, so all λ_j (and μ_i) are 0.

5. Count: $\dim V = m + n = \dim \operatorname{null} T + \dim \operatorname{range} T$.

□

Proposition (Prop. 25: equal dimensions)

If $\dim V = \dim W < \infty$ and $T \in \mathcal{L}(V, W)$, then T injective $\iff$ T surjective $\iff$ T bijective. In particular this holds for every operator on a finite-dimensional space.

The idea

This is the pigeonhole principle (W5) with dimension in place of size. Rank–nullity is the bookkeeping: every dimension lost to the null space is a dimension missing from the range.

Proof. 1. If T is injective, $\dim \operatorname{null} T = 0$, so $\dim \operatorname{range} T = \dim V = \dim W$, so $\operatorname{range} T = W$ (W4).

2. If T is surjective, $\dim \operatorname{range} T = \dim W = \dim V$, so $\dim \operatorname{null} T = 0$ and T is injective.

□

- **Consequences of the count.** If $\dim V > \dim W$, no linear map $V \to W$ is injective ($\dim \operatorname{null} T \geq \dim V - \dim W > 0$). If $\dim V < \dim W$, none is surjective ($\dim \operatorname{range} T \leq \dim V$) (LADR 3.22, 3.24).
- **Homogeneous systems.** m linear equations in $n > m$ unknowns with right side 0 have a nonzero solution: the map $\mathbb{F}^n \to \mathbb{F}^m$ is not injective. A system of $m > n$ equations in n unknowns has no solution for some right side: the map is not surjective.
- **The formula is about dimensions only.** For $P(x, y) = (0, y)$: $\operatorname{null} P = \operatorname{span}(e_1)$, $\operatorname{range} P = \operatorname{span}(e_2)$, and $\mathbb{R}^2 = \operatorname{null} P \oplus \operatorname{range} P$. For $N(x, y) = (y, 0)$: $\operatorname{null} N = \operatorname{span}(e_1) = \operatorname{range} N$. Still $2 = 1 + 1$, but the null space and the range are the same line.

§2.3 Matrices (Ch. 6)

The question. A linear map is determined by what it does to a basis (Prop. 21). How do we record that as numbers, and what computation with the numbers corresponds to composing maps?

Definition (Def. 16, 17 / LADR 3.29, 3.31)**.** $\mathbb{F}^{m,n}$ is the set of $m \times n$ matrices $A = (a_{ij})$ (m rows, n columns) with entries in $\mathbb{F}$. Let $B = (v_1, \ldots, v_n)$ be a basis of V and $C = (w_1, \ldots, w_m)$ a basis of W. The **matrix of** $T \in \mathcal{L}(V, W)$ is $\mathcal{M}(T) = (a_{ij}) \in \mathbb{F}^{m,n}$ where

$$T(v_j) = a_{1j} w_1 + \cdots + a_{mj} w_m.$$

- **Column j of $\mathcal{M}(T)$ is the coordinate vector of $T(v_j)$ in the basis C.** Everything about matrices follows from this sentence.
- $\mathbb{F}^{m,n}$ is a vector space of dimension mn, with basis the matrices $E_{i,j}$ (a single 1 in row i, column j). Ex. 27: $E_{1,1}, E_{1,2}, E_{2,1}, E_{2,2}$ for $\mathbb{F}^{2,2}$.
- Ex. 26: addition and scalar multiples are entry by entry, e.g. $(1+i)\left(\begin{smallmatrix} 1+i & 4 \\ 5 & i \end{smallmatrix}\right) = \left(\begin{smallmatrix} 2i & 4+4i \\ 5+5i & -1+i \end{smallmatrix}\right)$.
- In the standard basis: $Pe_1 = 0$, $Pe_2 = e_2$ gives $\mathcal{M}(P) = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$. $R_\theta e_1 = (\cos\theta, \sin\theta)$, $R_\theta e_2 = (-\sin\theta, \cos\theta)$ gives $\mathcal{M}(R_\theta) = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$. $Ne_1 = 0$, $Ne_2 = e_1$ gives $\mathcal{M}(N) = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$.

Example (Ex. 28: building a matrix column by column)

$T\colon \mathcal{P}_3(\mathbb{C}) \to \mathbb{C}^4$, $T(p) = (p(1), p(i), p(-1), p(-i))$, bases $1, x, x^2, x^3$ and $e_1, \ldots, e_4$.

1. $T(1) = (1, 1, 1, 1)$; $T(x) = (1, i, -1, -i)$; $T(x^2) = (1, -1, 1, -1)$; $T(x^3) = (1, -i, -1, i)$.
2. These are the columns:

$$\mathcal{M}(T) = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}.$$

Proposition (Prop. 26)

With bases fixed, $T \mapsto \mathcal{M}(T)$ is a bijection $\mathcal{L}(V, W) \to \mathbb{F}^{m,n}$, and $\mathcal{M}(S+T) = \mathcal{M}(S) + \mathcal{M}(T)$, $\mathcal{M}(\lambda T) = \lambda \mathcal{M}(T)$. Hence $\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$.

Proof. 1. *Injective.* If $\mathcal{M}(S) = \mathcal{M}(T)$, then for each j the vectors Sv_j and Tv_j have the same coordinates in C, so $Sv_j = Tv_j$. Two linear maps that agree on a basis are equal (Prop. 21).

2. *Surjective.* Given any matrix $A = (a_{ij})$, let $w'_j = a_{1j} w_1 + \cdots + a_{mj} w_m$. By Prop. 21 there is a linear T with $Tv_j = w'_j$, and then $\mathcal{M}(T) = A$.

3. *Linear.* $(S+T)v_j = Sv_j + Tv_j$, so column j of $\mathcal{M}(S+T)$ is the sum of the j-th columns. The same for λT.

4. *Dimension.* A linear bijection sends a basis to a basis (it preserves both spanning and independence), so $\dim \mathcal{L}(V, W) = \dim \mathbb{F}^{m,n} = mn$.

□

Definition (Def. 18, 19 / LADR 3.73, 3.76: coordinates and matrix times vector)**.** The **coordinates** of $v = \lambda_1 v_1 + \cdots + \lambda_n v_n$ in the basis B are $\mathcal{M}(v) = (\lambda_1, \ldots, \lambda_n)^{\mathsf{T}}$. The product of a matrix and a column is defined so that

$$\mathcal{M}(T)\,\mathcal{M}(v) = \mathcal{M}(Tv).$$

In entries: if $A = (a_{ij})$ and $x = (\lambda_j)$, then $(Ax)_i = \sum_j a_{ij}\lambda_j$.

The idea

The product is not an arbitrary rule; it is what the coordinates of Tv have to be. $Tv = \sum_j \lambda_j T(v_j)$, and $T(v_j)$ has coordinates "column j". So the coordinates of Tv are $\lambda_1(\text{column } 1) + \cdots + \lambda_n(\text{column } n)$. Reading this row by row gives $(Ax)_i = \sum_j a_{ij}\lambda_j$.

So Ax is a combination of the columns of A with the entries of x as coefficients. Two consequences: the range of A is the span of its columns, and Ae_j is column j of A.

- Ex. 29: in $\mathcal{P}_2(\mathbb{R})$, $1 + 2x + 3x^2$ has coordinates $(1, 2, 3)$ in the basis $1, x, x^2$ and $(0, 1, 2)$ in the basis $1 + x,\ 1 + x^2,\ x + x^2$, because $0(1 + x) + 1(1 + x^2) + 2(x + x^2) = 1 + 2x + 3x^2$.
- Ex. 30:

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ 7 \\ 9 \end{pmatrix} = \begin{pmatrix} 5 + 14 + 27 \\ 10 + 21 + 36 \end{pmatrix} = \begin{pmatrix} 46 \\ 67 \end{pmatrix}.$$

- Ex. 31: for the T of Ex. 28, $T(1 + 2x + x^2)$: multiply $\mathcal{M}(T)$ by $(1, 2, 1, 0)^{\mathsf{T}}$ to get $(4, 2i, 0, -2i)$. Check: $p(1) = 4$, $p(i) = 1 + 2i - 1 = 2i$.

Definition (Def. 20 / LADR 3.43: matrix product)**.** For $T \in \mathcal{L}(U, V)$, $S \in \mathcal{L}(V, W)$ and bases of U, V, W: the product is defined so that $\mathcal{M}(S)\mathcal{M}(T) = \mathcal{M}(ST)$. In entries, if $A = (a_{ij})$ is $n \times m$ and $B = (b_{jk})$ is $m \times l$, then $(AB)_{ik} = \sum_{j=1}^{m} a_{ij}b_{jk}$.

The idea

Column k of $\mathcal{M}(ST)$ is the coordinates of $S(T(u_k))$. The coordinates of $T(u_k)$ are column k of B, and applying S to a vector multiplies its coordinates by A. So *column k of AB is A times column k of B*. Writing this out gives $\sum_j a_{ij}b_{jk}$. This is why the number of columns of A must equal the number of rows of B: both are $\dim V$.

Derivation of the entry formula.

$$ST(u_k) = S\Big(\sum_j b_{jk}v_j\Big) = \sum_j b_{jk}S(v_j) = \sum_j b_{jk}\sum_i a_{ij}w_i = \sum_i \Big(\sum_j a_{ij}b_{jk}\Big)w_i.$$

The coefficient of w_i is the (i, k) entry of $\mathcal{M}(ST)$. □

- Ex. 32: T = differentiation $\mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$, S = antiderivative with zero constant term $\mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$:

$$\mathcal{M}(S)\mathcal{M}(T) = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/3 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}:$$

  ST removes the constant term.
- Prop. 27: $(AB)C = A(BC)$, $A(B + C) = AB + AC$, $(A + B)C = AC + BC$, $AI = A$, $IB = B$. Each holds because the corresponding statement holds for compositions of maps; no computation with indices is needed.
- **Permutation matrices.** For a permutation σ of $\{1, \ldots, n\}$, $S_\sigma e_j = e_{\sigma(j)}$. Its matrix has exactly one 1 in each row and column. $S_\sigma^k = S_{\sigma^k}$.
- **Vectors as representations** (notes' remark): in machine learning a token is a vector $x \in \mathbb{R}^d$ and a learned matrix W acts by $x \mapsto Wx$. This is an ordinary linear map; the dimension is just large.

Stochastic matrices and Markov chains (Ch. 6.3).

- A **probability vector** $p \in \mathbb{R}^n$ has $p_j \geq 0$ and $\sum p_j = 1$; p_j is the probability of being in state j.
- $P \in \mathbb{R}^{n,n}$ is **stochastic** if every entry is ≥ 0 and every *column* sums to 1. Column j is the distribution of the next state, given the present state is j (it is Pe_j).
- If the present distribution is p, the next one is Pp: $(Pp)_i = \sum_j p_{ij}p_j$ adds, over present states j, (probability of being in j) $\times$ (probability of moving from j to i).
- Pp is again a probability vector: entries ≥ 0, and

$$\sum_i (Pp)_i = \sum_j (\sum_i p_{ij}) p_j = \sum_j p_j = 1.$$

- A **Markov chain** applies the same P repeatedly: $p_k = P^k p_0$. Other books use row vectors and row-stochastic matrices; that is the transpose of this convention.
- A permutation matrix is stochastic: from state j the next state is certainly $\sigma(j)$.

Example (Ex. 33)

Take the stochastic matrix

$$P = \begin{pmatrix} 1/2 & 1/4 & 0 \\ 1/2 & 1/2 & 1/2 \\ 0 & 1/4 & 1/2 \end{pmatrix}$$

and start from $p_0 = (1, 0, 0)$. The next three distributions are

$$\begin{aligned} p_1 &= Pp_0 = (\tfrac{1}{2}, \tfrac{1}{2}, 0) \qquad \text{(the first column of } P\text{)}, \\ p_2 &= Pp_1 = (\tfrac{3}{8}, \tfrac{1}{2}, \tfrac{1}{8}), \\ p_3 &= Pp_2 = (\tfrac{5}{16}, \tfrac{1}{2}, \tfrac{3}{16}). \end{aligned}$$

They approach $\pi = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$, and $P\pi = \pi$. Week 7 explains the limit and the speed using eigenvectors.

Shortcuts

- **Matrix of a map**: apply T to each basis vector of the domain, write the result in the basis of the target, and use it as a column.
- **Range**: spanned by the columns (the images of the basis). **Null space**: solve $Ax = 0$.
- **Rank–nullity does half the work**: once you know one of $\dim \operatorname{null} T$, $\dim \operatorname{range} T$, the other is $\dim V$ minus it. Find whichever is easier.
- **Dimension of a subspace cut out by conditions**: write it as $\operatorname{null} S$ for a map S to $\mathbb{F}^k$; if S is surjective, the dimension is $\dim V - k$ (HW3 P3).
- **Equal dimensions**: check only injectivity (solve $Tv = 0$) to get bijectivity.
- **Impossibility by counting**: $\operatorname{null} T = \operatorname{range} T$ forces $\dim V = 2 \dim \operatorname{null} T$, so $\dim V$ is even (HW3 P2).
- **Defining a map**: say where a basis goes (Prop. 21). To show two linear maps are equal, check them on a basis.
- **Not linear?** Check $T(0) = 0$ first.
- **Over** $\mathbb{F}_2$: a system with solution space of dimension k has 2^k solutions (HW3 P6, P7).

Traps

- $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$ does not mean $V = \operatorname{null} T \oplus \operatorname{range} T$: for N, they are equal. (They are not even subspaces of the same space unless $W = V$.)

- Independent lists need not stay independent under T (take $T = 0$); dependent lists always stay dependent (HW3 P4). Independence is preserved exactly when T is injective.
- "Injective $\iff$ surjective" needs $\dim V = \dim W$ finite. The shift S on sequences is surjective but not injective.
- $ST \neq TS$ in general, and $ST = 0$ does not give $S = 0$ or $T = 0$: $N^2 = 0$.
- The matrix depends on the bases. The same map has different matrices in different bases.
- Stochastic here means columns sum to 1, not rows.

§3 Problem set

HW3 P1. Let $T\colon V \to W$ be linear, V finite-dimensional, U a subspace of W.

(a) Prove that $T^{-1}(U) = \{v \in V : Tv \in U\}$ is a subspace of V.

(b) Prove $\dim T^{-1}(U) = \dim \operatorname{null} T + \dim(U \cap \operatorname{range} T)$.

(c) Let D be differentiation on $\mathcal{P}_4(\mathbb{R})$. Using (b), find the dimension of the space of p with $p'(x) = ax^2 + b$ for some $a, b \in \mathbb{R}$.

Hint: apply rank–nullity to the restriction of T to $T^{-1}(U)$.

HW3 P2. (a) Find a linear map $T\colon \mathbb{R}^4 \to \mathbb{R}^4$ with $\operatorname{null} T = \operatorname{range} T$. (b) Prove that no linear map $T\colon \mathbb{R}^5 \to \mathbb{R}^5$ satisfies $\operatorname{null} T = \operatorname{range} T$.

Hint: rank–nullity; model (a) on N.

HW3 P3. Let $U = \{p \in \mathcal{P}_4(\mathbb{R}) : p(1) = p'(1) = 0\}$. Define a linear map $S\colon \mathcal{P}_4(\mathbb{R}) \to \mathbb{R}^2$ with null space U and use rank–nullity to find $\dim U$. Compare with Homework 2.

Hint: $S(p) = (p(1), p'(1))$; show it is surjective.

HW3 P4. Let $T\colon V \to W$ be linear. (a) Prove that if $v_1, \dots, v_n$ is dependent, so is $Tv_1, \dots, Tv_n$. (b) Give an example where an independent list is not sent to an independent list.

Hint: apply T to a relation.

HW3 P5. Define $T\colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_4(\mathbb{R})$ by $(Tp)(x) = x^2 p'(x) + p(x)$. (a) Find the matrix of T in the bases $1, x, x^2, x^3$ and $1, x, x^2, x^3, x^4$. (b) Determine $\operatorname{null} T$ and $\operatorname{range} T$ and their dimensions. (c) Explain from (a) why the range is the span of the columns.

Hint: column j is the image of the j-th basis vector.

HW3 P6. Solve $x + y + z + w = 1$, $x + z = 0$ over $\mathbb{F}_2$. List all solutions.

Hint: $-1 = 1$ and $2 = 0$.

HW3 P7 (optional). Let A be an $n \times m$ matrix over $\mathbb{F}_2$ with $Ax = b$ consistent. Express the number of solutions in terms of m and $\operatorname{rank} A$.

Hint: the solutions are one solution plus the null space.

R1 (Practice Midterm 1, Question 3). Define $T\colon \mathbb{R}^3 \to \mathcal{P}_2(\mathbb{R})$ by $T(a, b, c) = (a + b) + (b + c)x + (a + c)x^2$. Use the basis $B = (1, 0, 0), (1, 1, 0), (1, 1, 1)$ of the domain and $C = (1, x, x^2)$ of the codomain. (a) Find $\mathcal{M}(T)$. (b) Prove that T is bijective.

R2 (Practice Midterm 1, Question 4d). Let $D\colon \mathcal{P}_4(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$ be differentiation and $U = \operatorname{span}(1, x^2) \subseteq \mathcal{P}_3(\mathbb{R})$. Use HW3 P1(b) to find $\dim D^{-1}(U)$, and give a basis of $D^{-1}(U)$.

R3 (Midterm 1 review, Exercise 2). Let $T\colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$, $T(p) = p'' + p'$. Compute $\mathcal{M}(T)$ in the standard bases.

R4 (Midterm 1 review, Exercise 6). Does there exist a linear map $T\colon \mathbb{R}^4 \to \mathbb{R}^2$ with $\operatorname{null} T = \{(x, x, x, x) : x \in \mathbb{R}\}$?